



Global greenhouse gas cobenefits of US irrigated agriculture

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Land use and land conversion for agriculture contribute nearly a quarter of global greenhouse gas (GHG) emissions, making the sector a key priority for emissions reductions. Crop yields are one factor that modulate agricultural land demand, with, all else equal, global land use change increasing with yield declines and decreasing with yield gains. Therefore, agricultural practices drive both direct, on-farm emissions, and indirect land use change emissions through yield effects. Here, we couple estimates of direct emissions from US irrigation with an assessment of its productivity impacts and associated avoided indirect land use change. We find that US irrigation increases crop production by 3 to 39% across crop types, providing at least USD \$31.9 billion in additional revenue. Using a spatially explicit global economic model, we project that these productivity benefits avoid net conversion of forests (−32 Mha) and other natural vegetation (−7 Mha) to grasslands (+35 Mha) and croplands (+4 Mha) globally. Due to land sparing, US irrigation provides an estimated gross GHG benefit of 6.86 Gt CO₂e, while producing 0.019 Gt CO₂e of direct emissions annually. Our findings highlight the necessity of jointly accounting for direct and indirect emissions to guide effective agricultural emissions mitigation policy.

irrigation | greenhouse gas emissions | land use change | yield modeling | land sparing

Agricultural systems face the dual challenge of reducing greenhouse gas (GHG) emissions (1–3) while also increasing crop production under increasingly adverse growing conditions (4, 5). Addressing these objectives concurrently is critical given that crop yields and agricultural GHG emissions are inextricably linked through land use and land use change (6), which currently account for ~12% of total CO₂ emissions (2, 7). At the global scale, increasing yields reduces demand for agricultural land, in turn limiting conversion of intact ecosystems to agricultural production (8–13). Conversely, practices that reduce yields or policies that increase crop demand lead to expansion of agricultural lands globally, producing emissions from land use change (14). Net GHG impacts of agricultural management practices are therefore a function of both direct, local emissions as well as emissions induced via indirect land use change associated with changes in yield (Fig. 1). Despite the importance of both direct emissions (i.e., from on-farm management) and indirect emissions (i.e., from yield-mediated effects on global agricultural land use), they are rarely jointly quantified. Recent work that assessed both dimensions in the context of cover cropping found that a large proportion of the carbon sequestration benefits are offset by indirect land use change associated with productivity losses, demonstrating the importance of joint consideration of direct and indirect emissions impacts (14). As interest and investment in climate-smart agriculture grows among policymakers and practitioners, comprehensive quantification of net emissions impacts is essential for guiding decision-making.

In this study, we evaluate the net GHG impacts of US irrigation, demonstrating a transferable model pipeline for comprehensively assessing the GHG impacts of any agricultural management practice via local and global pathways. Irrigation is a widespread and highly impactful management practice, installed on approximately 20% of global croplands and estimated to account for roughly 40% of global food production and 70% of water withdrawals (15), though these estimates vary across sources (16, 17) and remain highly uncertain (18, 19). Expansion of irrigation, which increases yields and buffers against increasing temperatures and precipitation variability, has been promoted as a potential climate change adaptation strategy (20–25). However, irrigation also produces GHG emissions through energy use, increased denitrification, and degassing of groundwater (26–30), with energy use being the dominant emissions pathway. Irrigation expansion for the sake of climate change adaptation can be undertaken alongside efforts to reduce irrigation emissions, such as incentives for electric pump adoption, electrical grid decarbonization, and reduced groundwater extraction (26–28). Despite local emissions, irrigation has the potential to reduce demand for agricultural land use by increasing yields, but these indirect emissions benefits remain unquantified. Low-emissions expansion of

Significance

Agriculture produces greenhouse gas emissions from on-farm activities and through conversion of natural ecosystems to agricultural use. Agricultural practices that increase yields can reduce demand for agricultural land, reducing emissions from land use change, but these indirect emissions impacts are rarely considered. We use US irrigation, which is widespread and highly impactful, as an exemplary practice for developing an analytical approach to net emissions accounting for agricultural management. This work provides estimates of the yield benefits of irrigation and reveals that the emissions benefits of avoided land use change outweigh annual, on-farm irrigation emissions by a factor of 363. It is therefore critical to quantify both on-farm emissions and indirect land use change impacts to inform climate-smart agricultural policy.

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Agricultural Management Practice (e.g., irrigation, cover crops, tillage, etc.)

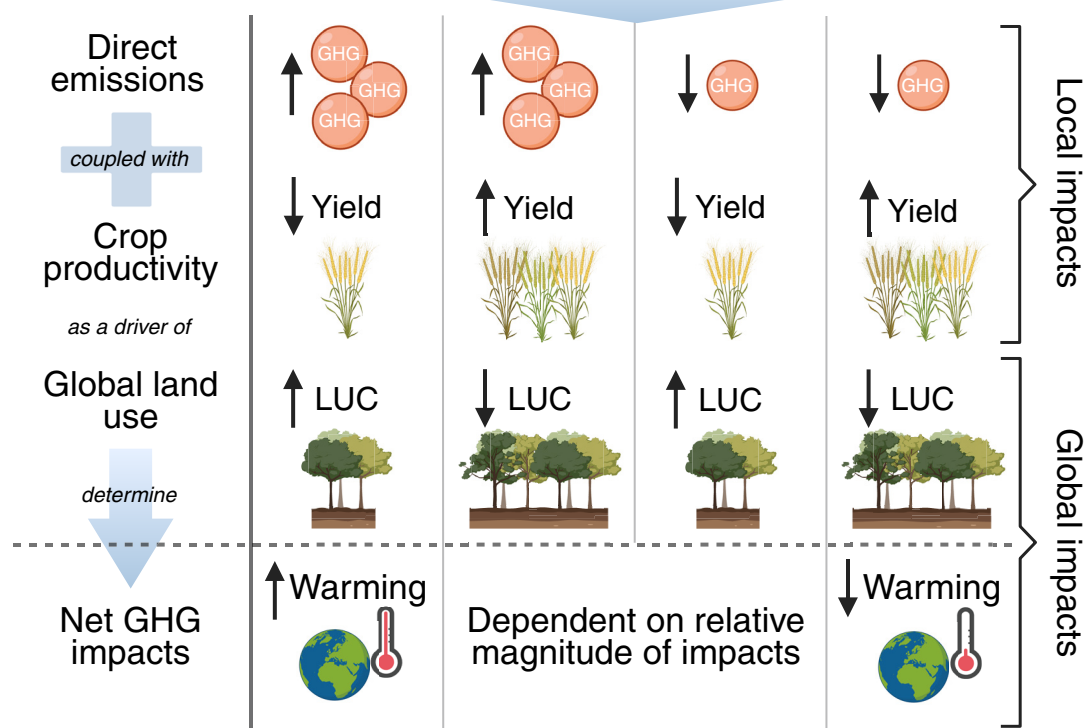


Fig. 1. Local and global mechanisms by which agricultural management practices drive net GHG emissions. Management practices are associated with direct, on-farm GHG emissions (first row). They also influence crop productivity (second row), which directly drives indirect emissions from global land use change (third row). Together, these local and global effects drive net GHG impacts (fourth row). Practices that increase local emissions and decrease yields lead to net increases in emissions (*Left* column). The net effects of practices that either increase both local emissions and yields (second column) or decrease both (third column) are dependent on the relative magnitude of each impact. Practices that decrease direct emissions and increase yields lead to net decreases in emissions (*Right* column).

irrigation in areas that will experience substantial yield gains may therefore provide an opportunity for synergistic achievement of climate change adaptation and mitigation goals in the agricultural sector, but comprehensive, net emissions accounting is needed to validate and quantify this potential.

Data and methodological limitations have historically constrained estimation of indirect land use change, defined as a shift in global land use induced by market responses to regional policies or practices (31), and associated GHG emissions (32). The economic and policy mechanisms that influence the extent and location of indirect land use change are complex, and the land sparing effects of yield increases remain an active area of research (9, 33, 34). However, recent advances across disciplines now enable quantification of the indirect land use change emissions associated with irrigation via three key computational steps: 1) empirically quantifying the production increase attributable to irrigation without reliance on process-based models, 2) projecting land use change that may occur in the absence of that additional production via spatially explicit Earth-Economy models with physical representation of land, and 3) estimating GHG emissions associated with projected land use change using high-resolution carbon stock data instead of coarse plant functional type-level estimates. Here, we integrate these model and data advances to provide a comprehensive assessment of the production, land use, and net GHG impacts of irrigation. The approach we develop for US irrigation is transferable, enabling comprehensive emissions accounting in support

of targeted mitigation actions in other regions and for other practices.

Results

Impacts of Irrigation on US Crop Production. To calculate the production benefits of US irrigation, we first modeled the ratio of rainfed to irrigated crop yields using historical, colocated yields from the United States Department of Agriculture (USDA) with weather and soil data, crop type, and year as predictors (see *Methods* for details). This model allows us to map the relative increase in yields attributable to irrigation across the country, providing the first spatially resolved, empirically based estimates of irrigation's yield benefits for the United States and revealing heterogeneous impacts across space (Fig. 2 and *SI Appendix, Fig. S1*) and crop types (*SI Appendix, Fig. S2*). Across the currently irrigated crop geographic distributions, our models indicate that irrigation increases yields by 35% (tobacco) to 121% (dry beans) (*SI Appendix, Fig. S3*), with crop-level variation dependent more on the geographic distributions of the crop than variation in intrinsic crop water demand (*SI Appendix, Fig. S2*). Our models predict historical yield ratios reasonably well (*SI Appendix, Fig. S4*; $RMSE = 0.16$, $r^2 = 0.53$ for the crop-agnostic model and $RMSE = 0.14$, $r^2 = 0.79$ for the crop-specific model), with vapor pressure deficit (VPD), actual evapotranspiration (AET), and crop type among the most influential predictors (*SI Appendix, Fig. S5*).

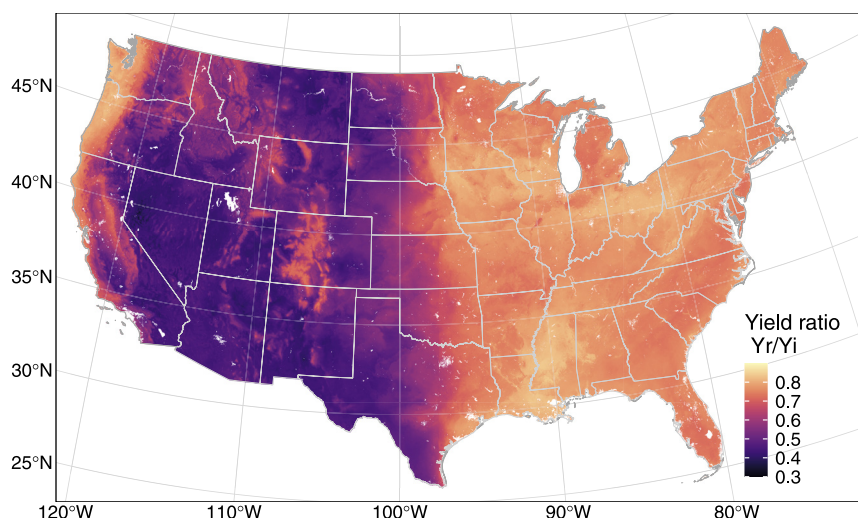


Fig. 2. Predicted ratios of rainfed to irrigated crop yields. Crop-agnostic yield ratios predicted using a random forest model trained on paired county-level rainfed and irrigated yield observations ($n = 49,217$) from USDA survey data with weather and soil predictors ($1/120^\circ$ resolution).

By coupling our predicted yield ratios with maps of crop-specific irrigated areas (35, 36), we estimate the hypothetical decline in national crop production under a zero-irrigation scenario in which we assume that all irrigated land is converted to rainfed production of the same crop (Fig. 3). This hypothetical scenario provides an estimate of irrigation's production benefits in their entirety while holding other factors constant, including cropland extents and crop type, though our yield ratio estimates also enable future analysis of alternative scenarios, such as crop-switching and localized contraction or expansion of irrigation.

Under the zero-irrigation scenario, we find that crop production declines by 3 to 39% across crop types, with an average production loss of 16% for all crops, weighted by irrigated area. 70% of the across-crop variability in production declines is determined by the proportion of crop area that is irrigated ($P < 0.001$; *SI Appendix, Fig. S6*), with the largest declines occurring for heavily irrigated crop groups such as fruits and vegetables and rice. Among crops with comparable proportions of irrigated area, production losses are largely controlled by the yield ratio. For example, barley, cotton, and peanuts exhibit different production losses (26%, 20%, and 14%, respectively) despite similar irrigated areas (33%, 36%, and 38%, respectively) due to their divergent rainfed:irrigated yield ratios (0.51, 0.60, and 0.69, respectively).

Impacts of these losses on agricultural revenues would be substantial. Based on national crop revenue reported in the 2017 Census of Agriculture, our estimates suggest that direct losses of USD \$31.9 billion would occur without irrigation (95% CI: USD \$22.7 to 42.4 billion; *SI Appendix, Fig. S7*). This estimate is conservative as it does not include potential irrigated land retirement nor economic losses from sugar crops, for which national-scale revenues were not reported in 2017. Losses are dominated by a USD \$18.6 billion loss in revenue from fruit and vegetable production, followed by a USD \$6.9 billion loss in maize production. The USDA estimated that costs for on-farm irrigation infrastructure, energy use, labor, and off-farm water purchases totaled USD \$9.5 billion in 2023, suggesting a conservative, private benefit-cost ratio of 3.4, or a net benefit of USD \$22.4 billion (37).

Land Sparing Attributable to Irrigation. Production losses were calculated within 18 agroecological zones (AEZs) for 16 USDA-based crop classes, aggregated to the eight crop groups represented in the Global Trade Analysis Project model (GTAP; *SI Appendix, Fig. S8 and Table S1*), and used as inputs to GTAP-AEZ-SEALS

(Spatial Economic Allocation Landscape Simulator) (38). The GTAP-AEZ-SEALS model is a widely used database and general equilibrium model that can simulate changes to the global economy and land use in response to economic shocks, all else held constant. The GTAP-AEZ-SEALS model endogenously accounts for price shifts, demand elasticities, crop product substitution, trade responses, and compensatory intensification of capital inputs to agriculture, while holding other factors constant, such as trade policy, agricultural technology, and exogenous drivers of crop product demand (i.e., biofuel policies and dietary preferences).

Using GTAP to simulate global land use change in response to the loss of irrigated production, we find that these US production losses trigger net global conversion of forest area (-31.7 Mha) and other natural vegetation (-6.9 Mha) alongside expansion of grasslands ($+34.8$ Mha) and croplands ($+3.8$ Mha) (Fig. 4A and *SI Appendix, Figs. S9 and S10*). Large gross changes underlie the projected net changes in land use, with global gross forest conversion of 35.8 Mha (95% CI: 23.1, 50.4 Mha) and gross expansion of 4.1 Mha (3.1, 5.0 Mha); gross grassland conversion of 5.7 Mha (3.4, 8.8 Mha) and gross expansion of 40.5 Mha (26.1, 53.7 Mha), gross conversion of other natural vegetation of 9.3 Mha

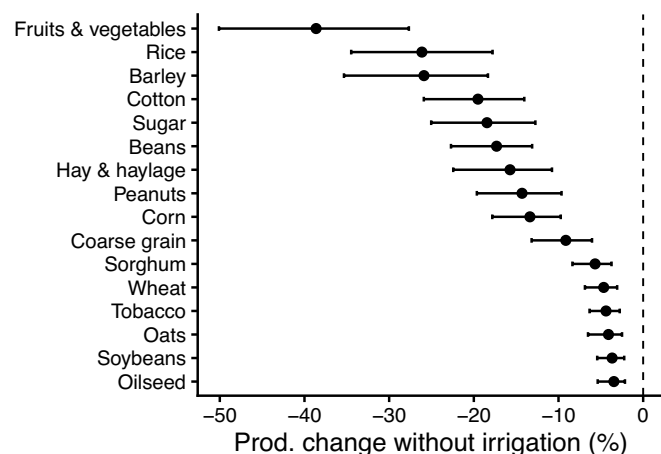


Fig. 3. Change in crop production in the United States under a hypothetical zero-irrigation scenario. Projected percent change in crop production under the assumption that all irrigated area is converted to rainfed production, calculated from modeled yield ratios and crop-specific irrigated areas. Error bars reflect the 95% CI for each estimate based on uncertainty in the yield ratio model.

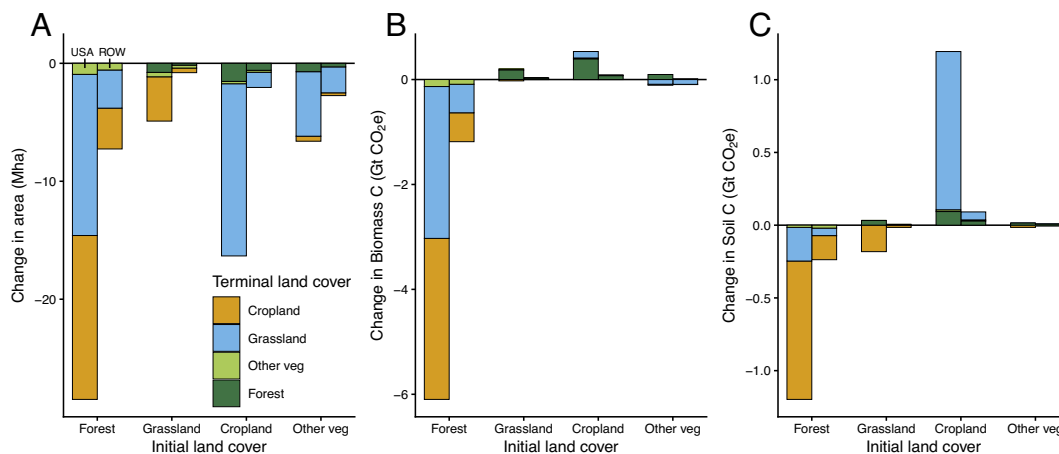


Fig. 4. Projected changes in global land cover and carbon stocks under the zero-irrigation scenario. Changes in (A) land cover (million hectares), (B) biomass carbon stocks (gigatons CO_2), and (C) soil carbon stocks (gigatons CO_2 ; 0 to 30 cm depth) in the United States (USA; Left bars) and the Rest of the World (ROW; Right bars) occurring for each initial land cover type based on GTAP-AEZ-SEALS projections under the zero-irrigation production shock. 95% CI for each bar are provided in [SI Appendix, Fig. S10](#).

(6.2, 12.0 Mha) and gross expansion of 2.5 Mha (1.5, 3.8 Mha), and gross cropland conversion of 18.4 Mha (11.8, 24.6 Mha) and gross expansion of 22.2 Mha (13.9, 33.3 Mha).

The gross changes in cropland area are strongly driven by redistribution of croplands from previously irrigated areas of the western United States to the US Northeast, Upper Midwest, and West Coast ([SI Appendix, Fig. S9](#)). In the model projections, lost cropland area in the western United States is primarily replaced with grazing lands. The large projected expansion in grasslands in the United States and the rest of the world compensates for declines in feed crop production on previously irrigated croplands. Notably, the expansion of both cropland and grazing lands primarily displaces forests, with some additional grazing land expansion into other natural vegetation classes ([Fig. 4A](#)).

Land use changes are strongly concentrated in the United States, which accounted for 80% of the gross conversion of forests, 86% of grasslands, 89% of croplands, and 71% of other natural vegetation ([Fig. 4A](#)). Net changes in land use are similarly dominated by the United States for forests, grasslands, and other natural vegetation, but croplands, in contrast, exhibit a slightly smaller net expansion in the United States (1.75 Mha) relative to the rest of the world (2.04 Mha). Other regions with large changes in land use include Mexico, which exhibits substantial conversion of forests and other vegetation to grazing lands and cropland, Canada, which exhibits grassland and cropland expansion, and Australia and New Zealand, which exhibit conversion of forests and other vegetation to grazing lands ([SI Appendix, Fig. S11](#)).

The projected land use change cannot be directly linked to individual crops, as the production losses across all crop groups were simulated as a simultaneous shock. We additionally ran crop-specific GTAP simulations, which were used only to allocate avoided emissions to the county level. Because of nonlinearities and interactions in economic responses, the sum of the indirect land use change projected from the crop-specific shocks is larger than from the total shock. While these crop-specific runs only approximate the relative influence of individual crops on the overall land use change, they suggest that the loss of fruit and vegetable production is responsible for the largest share of cropland expansion, perhaps because of limited substitutability of other agricultural products for fruits and vegetables, followed by coarse grains and then oilseeds.

Avoided Emissions from Land Sparing. We estimated the impacts of the projected land use change on ecosystem carbon stocks using spatially explicit datasets of carbon stocks in aboveground vegetation, belowground vegetation, and soil organic carbon (SOC). In total, we estimate that land sparing from US irrigation has avoided approximately 6.86 Gt CO_2 emissions (95% CI: 4.85 to 8.88 Gt; [Fig. 4B](#) and [SI Appendix, Figs. S10 and S12](#)). Changes to biomass carbon stocks under the projected land use change scenario would produce a net total of 6.55 Gt CO_2 emissions (95% CI: 5.22 to 7.88 Gt), including 7.06 Gt of CO_2 emissions from the clearing of forests for cropland and grazing land, sequestration of 0.67 Gt CO_2 on the small areas of croplands and grasslands converted to forests, and minor changes from other land cover transitions ([SI Appendix, Tables S2 and S3](#)). Consistent with the distribution of land use change, a large majority of the forest loss emissions from biomass occur in the United States (84%; [Fig. 4B](#) and [SI Appendix, Fig. S12](#)).

Changes in soil carbon stocks under the zero-irrigation scenario are smaller in magnitude than changes in biomass carbon stocks but are also subject to more complex and uncertain moderating factors. Globally, the projected loss of forests and conversion of grasslands and other vegetation to croplands is associated with gross soil carbon emissions of 1.65 Gt CO_2 to 30 cm depth (95% CI: 1.23 to 2.09 Gt CO_2 ; [Fig. 4C](#) and [SI Appendix, Figs. S10 and S13](#)). However, these carbon losses are largely offset by gains due to the replacement of croplands with grasslands (which increases carbon stocks by 1.15 Gt CO_2 ; 95% CI: 0.93 to 1.37) and the small increases in forest area (0.18 Gt CO_2) and other vegetation (0.02 Gt CO_2). Thus, we estimate that the projected land use change would result in net soil carbon losses of 0.31 Gt CO_2 (95% CI: -0.38 to 1.00 Gt CO_2). Underlying these estimated changes, our data suggest per-hectare losses of 17.6 t C from forest to cropland conversion, 13.0 t C from grassland to cropland conversion, and 4.6 t C from forest to grassland conversion, and per-hectare gains of 15.6 t C for cropland to forest conversion, 19.7 t C for cropland to grassland conversion, and 11.2 t C for grassland to forest conversion for the top 30 cm (see [SI Appendix, Table S2](#) for additional land use changes). These values depend on estimates of baseline soil carbon stocks, which vary across data sources (39), and by the geographic distribution of land use change, which influences the magnitude of soil carbon gain or loss under a given transition (40). The reported CIs incorporate

quantified uncertainty in both initial soil carbon stocks and stock changes under land use change.

Our estimates of soil carbon emissions are likely to be conservative for several reasons. First, land use change influences soil carbon beyond 30 cm depth (41, 42), but we only model changes in the 0 to 30 cm stock because of limited data availability and larger uncertainties for deeper soil carbon stocks (43). Second, reaching equilibrium soil carbon levels following land use change often takes >80 y, with losses typically proceeding more quickly than gains (40, 44, 45). The effect sizes leveraged to estimate soil carbon changes reflect a wide range of time since conversion (1 to 200 y) with an average of 20 y, suggesting that these estimates are unlikely to reflect a new equilibrium (40). Third, the projected expansion of grasslands is driven by expansion of grazing, but this analysis does not account for impacts of grazing on soil carbon, which are mixed (46–48). Fourth, we do not explicitly model increases in local soil carbon stocks due to irrigation, which are common in arid areas (49, 50). This may lead to an overestimate of soil carbon gain associated with the cropland to grassland transition, which likely in part reflects conversion of irrigated cropland to rainfed grazing land. Finally, additional uncertainty is introduced by the absence of historical land use in the models

underlying the initial soil carbon stock data, which may lead to misrepresentation of soil carbon in pixels that have undergone land use change in recent decades.

Coupling Irrigation GHG Benefits with Emissions. We estimate the net GHG impacts of irrigation by coupling avoided emissions from land sparing with estimates of GHG emissions from irrigation due to pump energy use, elevated N₂O, and groundwater CO₂ degassing. Amortized over a 20-year period, we estimate that irrigation produces 0.38 Gt CO₂e and avoids 6.86 Gt CO₂e (Fig. 5A), indicating that the GHG benefits of irrigation in the United States exceed annual irrigation-related GHG emissions by a factor of 363. Emissions from pump energy use comprise 72% of total GHG emissions from irrigation but are mitigatable via pump electrification and electrical grid decarbonization. Under a best-case scenario in which irrigation energy use is fully decarbonized, 20-y emissions would fall to just 0.11 Gt CO₂e, or 1.5% of the avoided emissions due to land sparing (SI Appendix, Fig. S14).

We used output from crop-specific GTAP shocks to attribute avoided emissions from land sparing to the county level based on the share of production increase occurring in each county for each crop group (Fig. 5B and SI Appendix, Figs. S14 and S15).

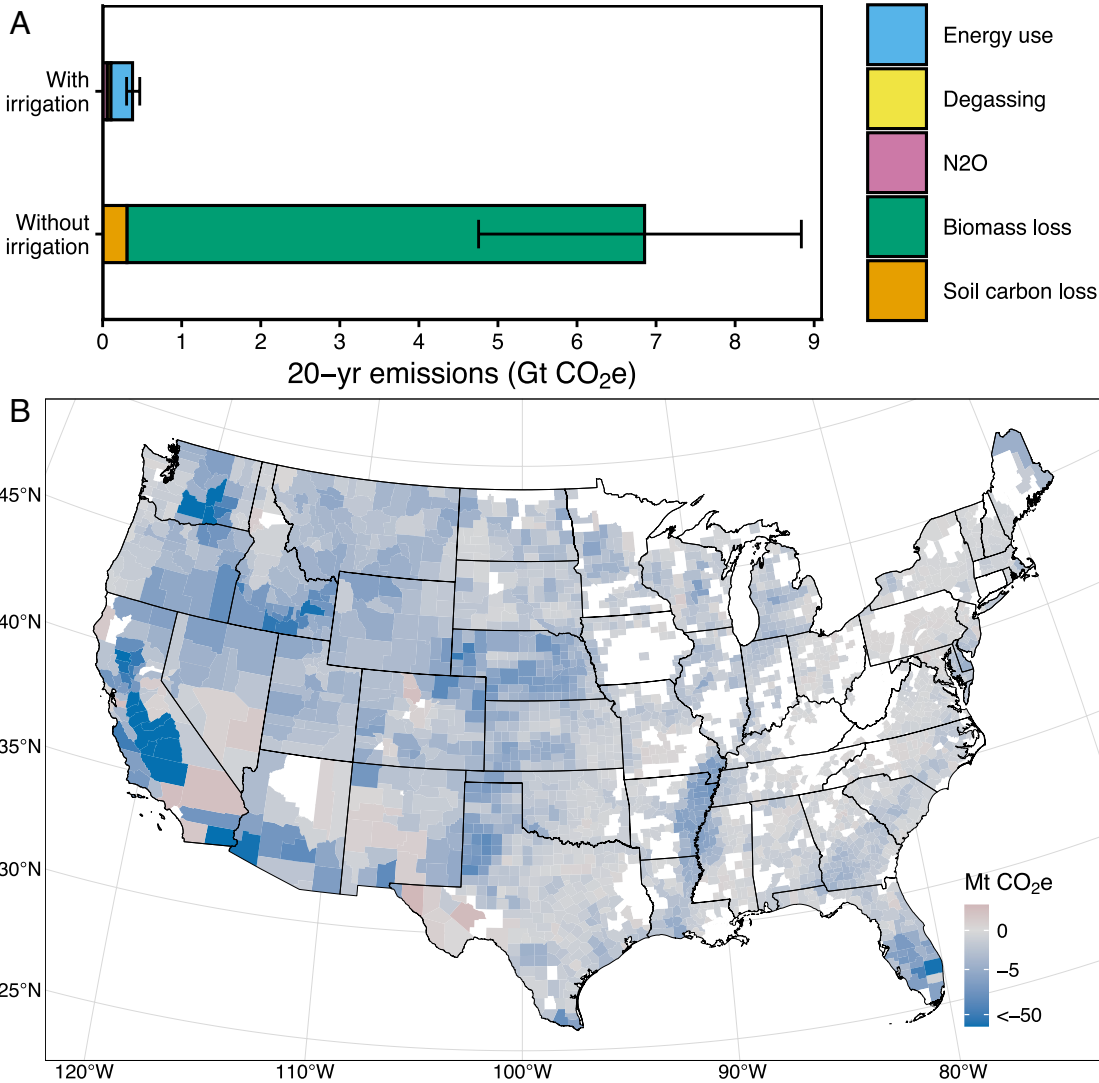


Fig. 5. Net GHG emissions from irrigation in the United States. (A) Total national impacts by emissions source under the current (“with irrigation”) and zero-irrigation (“without irrigation”) scenarios, and (B) net emissions downscaled to the county level, based on 20 y of direct emissions and the proportion of avoided emissions allocated based on the share of crop-specific relative production loss.

County-level shares of production losses are a function of irrigated area, irrigated to rainfed yield ratios, and irrigated yields, with the largest relative production losses occurring in places with large irrigated extents, low ratios of rainfed to irrigated yields, and high irrigated yields (*SI Appendix, Fig. S15*), and the crop-specific outputs were therefore used only to determine the spatial distribution of avoided emissions.

We estimate that irrigation provides a net GHG benefit in 2,141 counties, or 89.9% of the counties with irrigated area represented in this analysis (*Fig. 5B*). If pump energy use was fully decarbonized, irrigation would provide GHG benefits in 97% of counties. Moreover, in counties with net positive GHG emissions associated with irrigation, the magnitude of those emissions is very small. The gross GHG emissions from these counties are only 0.69 Mt CO₂e per year without pump decarbonization, representing less than 4% of the national gross annual irrigation-related GHG emissions. With decarbonization, gross emissions are only 0.02 Mt CO₂e per year. These counties tend to have small irrigated extents (containing only 0.1% of the total irrigated area represented in this study) and therefore small absolute productivity gains, while also relying on energy-intensive water sources.

This database of county-level net emissions impacts enables future analysis of the productivity and emissions impacts of various irrigation scenarios, such as localized expansion or contraction of irrigated area, groundwater depletion, crop switching, and changes in irrigation technology. However, county-level estimates of avoided emissions should be interpreted in the context of the large uncertainties associated with attributing global land use change to a localized production change, as well as the conservative calculation of production benefits. Land sparing outcomes of yield-increasing practices may also depend on whether demand-expanding policies, such as biofuel mandates, are adopted in response to increased production. In this study, we isolate the effects of irrigation by holding such policies constant.

Discussion

Climate-smart agricultural policies depend on accurate assessment of the GHG impacts of agricultural management practices. Although much work focuses on estimating either on-farm GHG emissions or productivity impacts of various management practices, complete characterization of GHG impacts requires rigorous quantification of both components coupled with an assessment of global land use change. Financial incentives for climate-smart practices are increasingly common, but typically provide compensation exclusively based on estimated direct emissions reductions or carbon sequestration, without consideration for indirect land use change emissions impacts. Inclusion of indirect emissions impacts can substantially change the magnitude (as in *ref. 14*) and even the directionality, as shown here, of the net emissions effects of a practice. Incorporating a comprehensive emissions accounting approach may therefore alter the types of practices incentivized by climate-smart agricultural policies and carbon markets. Here, we integrated three key methodological advances to provide a transferrable modeling framework for concurrent assessment of local emissions and indirect emissions from global land use change associated with productivity impacts.

First, irrigation's contribution to crop production remains quite uncertain, with little support for the oft-cited claim that irrigation is responsible for ~40% of global crop production (*18*). Large-scale estimates of the yield impacts of irrigation are sparse and have traditionally relied primarily on process-based models (*51–55*). By leveraging an extensive dataset of colocated irrigated and

rainfed yield observations, recent advances in irrigation and crop type mapping, and a machine learning model able to capture complex relationships between yield ratios and environmental drivers, we develop a fully empirical, high-resolution estimate of irrigation benefits for crop productivity in the United States, an approach not previously applied at this scale and resolution. Prior work has found that process-based models can overestimate yield benefits relative to empirical models (*56*). Direct comparisons between our yield benefit estimates and other commonly used datasets, including those from AQUASTAT (*16*) and from Siebert and Döll (*17*), are complicated by differing spatial scales, crop resolution, and data years. However, we also found that the weighted average yield ratio across all croplands from our empirical approach (0.68) is slightly higher than that estimated by AQUASTAT for the United States (0.63). Similarly, our estimate of the weighted average yield ratio on irrigated croplands (0.61) was substantially higher than that from Siebert and Döll for irrigated croplands in North America (0.46). Moreover, our highly resolved dataset reveals large geographic and crop-specific variability, with potential to advance work assessing the socioeconomic impacts of irrigation practices and policy. While this step of the modeling process is the most specific to irrigation, this approach can be used to evaluate the productivity effects of any practice for which the distribution of the practice is known and sufficient colocated data on yields under the practice of interest and a baseline are available.

Second, methods to assess indirect land use change have often been limited by simplistic representation of economic drivers and/or low spatial resolution of land use change projections (*8, 11, 12, 14, 57–60*). Specifically, some studies have relied on the assumption that any lost production is replaced at national or global average yields, without accounting for economic responses (such as price, demand, or trade shifts) or the spatial distribution of projected cropland changes (e.g., *refs. 11 and 12*). An alternative approach involves estimating land use change based on the elasticity of agricultural land use to changes in agricultural total factor productivity [TFP, the ratio of agricultural sector outputs to inputs (*14, 58*)]. This empirical approach has the benefit of capturing some economic responses while avoiding the complexity and assumptions required for general equilibrium models. However, the TFP-based approach is restricted to the national scale, does not differentiate between agricultural commodities or between changes to inputs relative to outputs, and cannot capture all economic responses, such as intensification of nonland inputs.

Recent developments in Earth-Economy modeling (*61*) have enhanced our ability to understand where economic activity drives land use change at high resolutions, accounting for shifts in prices, demand, trade, input intensity, and product substitution, using general equilibrium models (*62, 63*). These improved models incorporate information about land inputs for different agroecological-zones (*64, 65*) to endogenously calculate where natural lands are converted to agriculture in economic equilibrium (*66*). They also include physical representations of land and elasticities of substitution across regions and land covers, accounting for differential suitability for crop production. Thus, they are well suited for linking with empirically driven models to downscale regional land use change estimates to high spatial resolution (e.g., SEALS) (*67*). Ultimately, these improved models simulate comprehensive, endogenous economic responses to economic shocks, while also incorporating improved biophysical constraints on indirect land use change. The primary limitation of general equilibrium models is that their complexity reduces interpretability and requires more extensive assumptions. Further work is needed

to directly compare estimates of indirect land use change across modeling approaches.

Third, models of the carbon impacts of land use change are traditionally constrained by reliance on static carbon stock estimates at the level of plant functional types (e.g., temperate deciduous forest, tropical grassland, etc.). As with IPCC Tier 1 or 2 estimates of land use change emissions factors, these models do not capture the substantial spatial heterogeneity in carbon stocks within plant functional types (7, 68–71). Because GTAP-AEZ-SEALS provides high-resolution projections of land use change, we are able to leverage the best-available spatially explicit data on vegetation and soil carbon stocks to provide more precise estimates of projected land use change emissions (43, 72–74). While uncertainties remain in global maps of carbon stocks in soils and vegetation, our approach represents a substantial improvement over relying on spatially invariant stock estimates.

Using this modeling approach, we conservatively demonstrate that US irrigation provides a net GHG benefit of 6.5 Gt CO₂e over 20 years due to land sparing, after accounting for local emissions. Irrigation practices are likely to shift in coming decades, with contractions in areas facing surface water scarcity, and expansion of irrigation in other areas, as benefits grow under increasingly variable precipitation regimes and warmer temperatures. Irrigation decision-making must consider impacts on water availability for nonagricultural users, aquifer sustainability, in-stream flows, and freshwater ecosystems (75–77), which are often most acute at local and regional scales and may be critical in water-stressed areas even when climate benefits are large. The data developed in this study allow for improved consideration of crop productivity and GHG emissions impacts alongside other economic, social, and environmental factors when developing local irrigation policies. At the national scale, focusing exclusively on direct emissions suggests that irrigation should be reduced in the interest of emissions mitigation, but comprehensive emissions accounting instead indicates that irrigation provides substantial emissions benefits via productivity gains. Therefore, irrigation is a synergistic strategy for addressing both mitigation and adaptation needs in many areas, with further benefits realizable through reducing pumping emissions intensity.

Methods

In brief, we first calculate the production increase attributable to US irrigation using colocated observations of irrigated and rainfed yields to develop a model of the ratio of rainfed to irrigated yields. Next, we assess changes in global land use under a hypothetical loss of US irrigated production using a global economic model. Finally, we quantify the emissions impacts of this projected land use change and calculate the net emissions impacts of irrigation at the county scale. A visual overview of the analytical process, including uncertainty estimation, is available in *SI Appendix, Fig. S16*.

Production Benefits of Irrigation. We estimated county-level yield benefits of irrigation using colocated data on rainfed and irrigated yields from the USDA National Agricultural Statistics Service (NASS) (78), monthly weather data from TerraClimate (79), and soil texture and organic carbon data from SoilGrids 2.0 (43). First, rainfed and irrigated yield data were extracted from the NASS QuickStats database, and the ratios (*R*) of rainfed yields to irrigated yields were calculated for 10 major irrigated crops from 1980 to 2022 at the county level for crop-county-years in which both irrigated and rainfed yields were available. Both irrigated and rainfed yields were available for barley, dry beans, corn, cotton, hay, oats, peanuts, sorghum, soybeans, and wheat, for a total of 49,217 unique crop-county-years (Dataset S1).

The observed yield ratio data were used to train a random forest model (80, 81) with weather and soil data as predictors. See *SI Appendix, Supplementary Text* for additional details on the selection of a random forest model. Weather variables used as model predictors included total potential evapotranspiration

(PET), AET, and precipitation, and average soil moisture, VPD, and minimum and maximum temperatures from TerraClimate. The weather data were aggregated to cropland-weighted county averages to exclude the influence of pixels that do not contain croplands, such as mountainous areas. Specifically, TerraClimate pixels were weighted by the proportion of pixel area that contained cropland area based on the 2023 USDA National Cultivated Layer dataset (30 m resolution) (82), which indicates if a pixel was cultivated in at least 2 years between 2019 and 2023, or in 2023 alone. After aggregation to the county level, months in which the mean minimum temperature was less than 0 °C were excluded to approximate the potential growing season. Finally, the monthly weather data were aggregated to growing season sums (for PET, AET, and precipitation) or means (for soil moisture, VPD, and temperatures). County average, depth-weighted percent sand, and percent SOC by volume were extracted from the SoilGrids 2.0 database (43) for the top 30 cm and cropland-weighted using the same protocol as the weather data. Year was included in the model to account for technological trends that may impact *R*.

We fit one crop-specific model that included a categorical variable for crop type, allowing the algorithm to capture heterogeneity in responses between crops and generate specific county-year-crop predictions. We also fit one crop-agnostic model trained using all available data to establish a general county-year yield ratio prediction that was used for crops absent from the NASS data. Hyperparameter tuning was conducted using 10-fold cross-validation based on the average RMS error (RMSE) to select the optimal model configuration from 32 combinations of typical random forest hyperparameter values. Gridded predictions of *R* were made at 300 m resolution for the contiguous United States in 2017 using potential growing season TerraClimate (79) (4 km resolution) and depth-weighted SoilGrids 2.0 (43) (300 m resolution) data as inputs. Average RMSE values were 0.16 for the crop-agnostic model and 0.14 for the crop-specific model. The lower performance of the crop-agnostic model likely reflects its inability to capture crop-specific differences in sensitivity to moisture conditions. Statistical uncertainty in both models was quantified using pixel-level 95% CI derived via an infinitesimal jackknife approach (83), which were then propagated through the remainder of the analysis (*SI Appendix, Fig. S16*). The crop-agnostic model, which has higher quantified uncertainty, was used across 19.9% of irrigated area, while the crop-specific model was used for the remaining 80.1%. In addition to the quantified uncertainty, we lack validation data for the crops simulated by the crop-agnostic model, introducing a source of unquantified uncertainty for those crops. We also conducted a sensitivity analysis to evaluate the effects of spatial and temporal autocorrelation on model performance and production shock estimates (*SI Appendix, Fig. S17 and Supplementary Text*).

The hypothetical percent loss in crop production under the zero-irrigation scenario was calculated at the level of AEZs and GTAP crop groups for input to the GTAP model (*SI Appendix, Fig. S8*). We selected a zero-irrigation scenario, rather than intermediate scenarios of localized irrigation contraction or deficit irrigation, to assess the productivity and land sparing benefits of irrigation in their entirety. This is a conservative estimate that serves as a useful lower bound of irrigation's production benefits as it does not account for the likely retirement of irrigated lands on which rainfed production is not economically feasible. First, the irrigated area (*A_i*), rainfed area (*A_r*), and average yield ratio (*R*) for each crop and AEZ were calculated based on the intersection of the 2017 Cropland Data Layer (36) (CDL; 30 m resolution) and the 2017 Landsat-based Irrigation Dataset (35) (LANID; 30 m resolution). Because of pixel-level uncertainties in LANID and CDL, irrigated and rainfed areas were then scaled to sum to the national total 2017 USDA Census areas (78). Area and yield ratio data were aggregated to the eight crop classes represented in GTAP (rice, wheat, coarse grains, fruits and vegetables, oilseeds, sugar crops, fiber crops, and other; alignment of NASS and GTAP crop groups is detailed in *SI Appendix, Table S1*). The relative shock (*S_{c,z}*) for each crop group (*c*) and AEZ (*z*) was then calculated according to Eq. 1, the derivation of which is detailed in *SI Appendix, Supplementary Text*. A 95% CI for each shock estimate was calculated using the 2.5th and 97.5th percentile estimates of *R*, which were used to propagate uncertainty in production changes throughout the analysis.

$$S_{c,z} = \frac{A_{i,c,z} - \left(\frac{1}{R_{c,z}}\right) \times A_{r,c,z}}{A_{r,c,z} + \left(\frac{1}{R_{c,z}}\right) \times A_{i,c,z}} \quad [1]$$

Land Use Change Prediction Via GTAP-AEZ-SEALS. The calculated production losses under the zero-irrigation scenario were used as shocks to the GTAP-AEZ model. A total of 11 runs using different shocks were conducted, including the central, 2.5th percentile, and 97.5th percentile estimates of the total shock for all crops, and the central shock estimates for each GTAP crop group individually. GTAP-AEZ is a computable general equilibrium model representing global production, consumption, and trade that has been widely used and validated in the global trade and resource economics fields for assessing indirect land use change (33, 63, 84). A detailed description of the development and validation of the specific model version used in this analysis is available in Johnson et al. (67). We used a static version of the model that represents 50 regions (countries or groups of countries) divided into 18 AEZs, which constrain land substitution based on environmental suitability. This model was run for the baseline (unshocked) scenario and the zero-irrigation scenario using the 2017 reference year of the GTAP 11 database under a static comparative framework to eliminate the influence of temporally variable factors (i.e., population growth, climate change, etc.). These model runs were used to simulate the physical extent of land use for crops, grazing, and managed forest under each scenario with all else held constant.

Land use projections from the GTAP-AEZ simulations were downscaled in two steps. First, they were disaggregated to 30 km resolution by scaling land cover estimates from the Land Use Harmonization (LUH2) data under the SSP2 scenario to match the GTAP-AEZ projections of land cover (85). Second, the SEALS algorithm was used to downscale land cover projections to 300 m resolution based on biophysical and socioeconomic predictors (38). The difference between the land cover map under the shocked scenario relative to the baseline scenario represents the land use change attributable to the loss of irrigated production.

For pixels projected to undergo land use change, the initial land cover was determined using the 2017 European Space Agency Climate Change Initiative medium-resolution land cover (ESA CCI MRLC) data, which represents 22 land cover classes and is the baseline map used by the GTAP-AEZ-SEALS model. The terminal land cover was determined by the SEALS output, representing seven land cover classes that were associated with the ESA CCI classes as detailed in *SI Appendix, Table S4*. Land use change was summed to the GTAP region-AEZ level for each transition (from each ESA CCI class to each SEALS class). Total area changes were then multiplied by estimates of per-hectare changes in soil and biomass carbon stocks (described below) to calculate the carbon emissions that would occur under the projected land use change. These calculations were completed for the outputs of each simulation. Estimates of land use change under the 2.5th percentile and 97.5th percentile shocks were used for subsequent uncertainty analysis.

Carbon Impacts of Land Use Change. We estimated the impacts of the projected land use change on steady-state carbon stocks in aboveground biomass, belowground biomass, and soil organic matter to a depth of 30 cm. For SOC, we rely on estimates of current SOC stocks from SoilGrids 2.0 (43) and data on the percent change in SOC under land use change from a recent meta-analysis (40). To align with the land use change projections, we calculated the average SOC stock for each unique GTAP region-AEZ-land cover combination, using the 2015 version of the ESA CCI MRLC land cover data because it was the base land cover map used by the SoilGrids 2.0 model. We also calculated the SE of the SOC values within each GTAP-AEZ-land cover polygon.

We used estimates of the effect of land use change on SOC from Huang et al. (40) under six different land use transitions (between forest, cropland, and grassland) in four latitude zones (tropical, -23.4° to 23.4° ; subtropical, -35° to -23.4° and 23.4° to 35° ; temperate, -35° to -50° and 35° to 50° ; subarctic, $<-50^\circ$ and $>50^\circ$). These effect sizes are based on 5,980 observations from previous studies, representing 1 to 200 y since land use change (mean = 20 y) and soil depths ranging from 1 to 98 cm (mean = 22 cm), though their effect size estimates do not explicitly account for depth or time since land use change. The median, 2.5th, and 97.5th percentile effect size estimates for each land use transition were then averaged to the GTAP-AEZ-land cover polygons based on the 2015 ESA CCI MRLC data, and the SE of the effect size was calculated as the absolute value of the 97.5th percentile minus the 2.5th percentile, divided by 3.92 (given that the CI is calculated as the mean $\pm$ 1.96*SE).

The per-hectare change in SOC (ΔSOC) was estimated according to Eq. 2, where $SOC_{initial}$ represents the initial (current) SOC stock to a depth of 30 cm

from SoilGrids ($t\ ha^{-1}$), and E refers to the proportionate change in SOC from the relevant land use transition (u) and climate zone (v) (40). To estimate a 95% CI for ΔSOC , we used a Monte Carlo bootstrapping approach to randomly generate and sample from truncated normal distributions for SOC_i and normal distributions of $E_{u,v}$ based on the mean and SE (10,000 iterations).

$$\Delta SOC = E_{u,v} \times SOC_{initial} \quad [2]$$

We calculated total biomass carbon stocks by adding woody plant leaf biomass from Graven et al. (86) to the above and belowground biomass carbon estimates from Spawn et al. (73) (300 m resolution). The mean and SE of total biomass carbon estimates were calculated at the GTAP-AEZ-land cover level based on the 2010 ESA CCI land cover data, consistent with the base map used by Spawn et al. (73). The per-hectare change in biomass carbon was estimated as the difference between the biomass carbon under the initial land cover from Spawn et al. and the average biomass carbon for the projected terminal land cover at the GTAP-AEZ-land cover level. A 95% CI for the change in biomass carbon was calculated using a Monte Carlo bootstrapping approach, in which we randomly generated and sampled from truncated normal distributions of initial and terminal biomass carbon stocks based on the mean and SE (10,000 iterations). Uncertainties remain in maps of soil and biomass carbon stocks and in estimates of ΔSOC under land use change. We reduce uncertainty by conducting this analysis at the level of GTAP-AEZ-land cover polygons rather than the pixel level, and we capture quantified uncertainty in biomass stocks, SOC stocks, and ΔSOC with the bootstrapping approach. As carbon stock maps continue to be refined, updated data products can be easily integrated into this model.

County-Level Net GHG Emissions Impact of Irrigation. Using the SEALS output from the single-crop model runs, we allocated the global GHG benefits of avoided land use change to US counties based on the proportion of the national production benefit that occurred in each county for each GTAP crop group. For each crop group (c), the proportion of production loss (L) occurring in each county (t) was calculated according to Eq. 3, using county-level irrigated area ($A_{i,ct}$), modeled ratios of rainfed to irrigated yields ($R_{c,t}$), and a modeled index of irrigated yields ($I_{c,t}$). The numerator of Eq. 3 represents production loss in county t , and the denominator represents the sum of production loss across all counties from $j = 1, \dots, N$. The irrigated yield index was calculated as the ratio of county-level irrigated yields to national irrigated yields, and it was modeled using data and methods identical to those used to model the rainfed to irrigated yield ratios. The county-level means of I and R were calculated over irrigated pixels for each crop group. The avoided emissions predicted from the crop-specific GTAP-AEZ-SEALS runs were scaled to match the avoided emissions from the complete shock, proportionately allocated to counties based on $L_{c,t}$ and then summed across crop groups.

$$L_{c,t} = \frac{I_{c,t} \times A_{i,ct} \times (R_{c,t} - 1)}{\sum_{j=1}^N I_{c,j} \times A_{i,cj} \times (R_{c,j} - 1)} \quad [3]$$

To account for direct emissions from irrigation, we use recently developed estimates of GHG emissions from irrigation energy use, elevated N_2O , and CO_2 degassing from groundwater (28, 87). Emissions from on-farm energy use were estimated using data on pump energy expenditures and energy source-specific prices and emissions factors (87). Emissions from energy use for interbasin transfers were estimated using either energy use or expenditure data reported by the operators of individual transfers. The effect of irrigation on N_2O emissions was estimated using a statistical model of N_2O emissions estimates that were produced for the US GHG Inventory with the DayCent ecosystem biogeochemical model (88, 89). Finally, emissions attributable to groundwater degassing of CO_2 were calculated using USGS data on groundwater extractions for irrigation and estimates of the partial pressure of CO_2 in groundwater based on measured groundwater pH and salinity (90).

Data, Materials, and Software Availability. All source data used in this manuscript are publicly available, though access to the GTAP 11 database requires purchase of a license. All code used for the analysis of

yield benefits and land use change emissions is available on Zenodo (<https://doi.org/10.5281/zenodo.16905434>) (91). The GTAP-AEZ-SEALS model code is available on Zenodo (<https://doi.org/10.5281/zenodo.7879386>) (92), though purchase of a GEMPACK license is required for running the GTAP model.

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